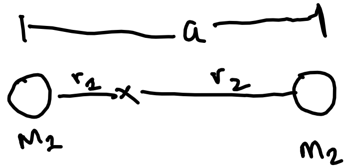


$$M_T = M_1 + M_2$$

$$q_1 = M_1 / M_2 \; ; \; q_2 = M_2 / M_1$$

$$\mu = \frac{M_1 M_2}{M_T}$$

$$r_1 = \frac{M_2}{M_T} a = \frac{M_2}{M_1 + M_2} a$$

$\longrightarrow \dfrac{r_1}{M_2} = \dfrac{a}{M_T}$; can show the same for $\dfrac{r_2}{M_1}$

$$\longrightarrow r_1 M_1 = \mu a = r_2 M_2$$

Can also then write: $q_1 = \dfrac{M_1}{M_2} = \dfrac{r_2}{r_1}$

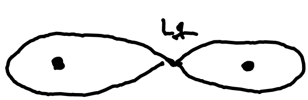
Kepler's third Law:

$$w^2 = \left(\frac{2\pi}{P}\right)^2 = \frac{G(M_1 + M_2)}{a^3}$$

Roche Potential in corotating frame:

$$\Phi_{Roche} = -\frac{GM_1}{|\vec{r} - \vec{r}_1|} - \frac{GM_2}{|\vec{r} - \vec{r}_2|} - \frac{1}{2}(\vec{\omega} \times \vec{r})$$

$\vec{F}_m = -\nabla \Phi_{Roche}\, m \longrightarrow$ net force on a test mass



Roche radius : $\frac{R_{RL,1}}{a} \simeq \frac{0.49\, q_1^{2/3}}{0.6\, q_1^{2/3} + \ln(1+q_1^{1/3})}$

$\longrightarrow$ if $R_1 > R_{RL,1}$ a star will fill its RL

$\longrightarrow$ you can use KEPLER III to calculate the orbital period at which a star of radius $R_\cdot$ and companion m_2 will fill its Roche Lobe.

Every orbit has an orbital energy and angular momentum :

energy: $E_{orb} = E_{pot} + E_{kin} = -\frac{G m_1 m_2}{a} + \frac{G m_1 m_2}{2a} = -\frac{G m_1 m_2}{2a}$

$$\left\{ E_{kin} = \frac{1}{2} m v_{orb}^2 \;;\; \text{but } F_{grav} = \frac{G m_1 m_2}{a^2} = \frac{m v_{orb}^2}{a} \quad \longrightarrow E_{kin} = \frac{1}{2}\frac{G m_1 m_2}{a} \right\}$$

angular momentum: $J_{orb} = J_1 + J_2 = M_1 r_1^2 \omega + M_2 r_2^2 \omega$

$$= \left[M_1 \left(\frac{M_2}{M_T} a\right)^2 + M_2 \left(\frac{M_1}{M_T} a\right)^2 \right] \omega$$

$$\longrightarrow J_{orb} = \frac{(m_1 m_2^2 + m_2 m_1^2)}{(M_1 + M_2)(M_1 + M_2)} a^2 \omega = \frac{m_1 m_2 (m_2 + m_1) a^2 \omega}{(m_1 + m_2)(m_1 + m_2)}$$

$$J_{orb} = \frac{M_1 M_2}{M_T} a^2 \omega = \mu a^2 \omega$$

can rewrite w/ KEPLER III:

$$J = \frac{M_1 M_2}{M_T} a^2 \omega = \frac{M_1 M_2}{M_T} a^2 \sqrt{\frac{G M_T}{a^3}} = M_1 M_2 \sqrt{\frac{G a}{M_T}}$$

Can also write J in terms of P_{orb}

$$J = \left(\frac{G^2}{2\pi}\right)^{1/3} \frac{M_1 M_2}{M_T^{1/3}} P^{1/3} = \sqrt{G}\, M_1 M_2 \sqrt{\frac{a}{M_T}}$$

ORBITAL EVOLUTION:

logarithmic derivatives: if $x = a y^b$, then

$$\frac{\dot{x}}{x} = \frac{b a y^{b-1} \dot{y}}{a y^b} = \frac{b \dot{y}}{y}$$

Then: $\dot{J} = \dot{J}_{M_1} + \dot{J}_{M_2} + \dot{J}_{M_T} + \dot{J}_P$

$$\longrightarrow \frac{\dot{J}}{J} = \frac{\dot{M}_1}{M} + \frac{\dot{M}_2}{M} - \frac{1}{3}\frac{\dot{M}_T}{M_T} + \frac{1}{3}\frac{\dot{P}}{P}$$

$$= \frac{\dot{M}_1}{M_1} + \frac{\dot{M}_2}{M_2} - \frac{1}{2}\frac{\dot{M}_T}{M_T} + \frac{1}{2}\frac{\dot{a}}{a}$$

* specific angular momentum: $h_i = \frac{J_i}{M_i} \rightarrow \dot{J}_i = h_i \dot{M}_i$

$\rightarrow J_i / J_{orb} = M_{(3-i)} / M_T \; ; \; h_i = J_{orb} M_{(3-i)} / M_i M_T$

We can use this equation to calculate orbital evolution: $\frac{\dot{a}}{a}$ for different conditions

Fast stellar winds : $v_{wind} >> v_{orb}$

if $\dot{M}_1 >> \dot{M}_2$ we can write:

$$\dot{M}_T = \dot{M}_1 \equiv \dot{M} \; ; \; \dot{M}_2 = 0 \, , \; \dot{J} = \dot{J}_1$$

$$\rightarrow \frac{1}{2}\frac{\dot{a}}{a} = \frac{\dot{J}}{J} - \frac{\dot{M}_1}{M_1} - \frac{\dot{M}_2}{M_2} + \frac{1}{2}\frac{\dot{M}_1}{M_T}$$

$$\rightarrow \frac{\dot{a}}{a} = \frac{2\dot{J}}{J} - \frac{2\dot{M}_1}{M_1} + \frac{\dot{M}_1}{M_T}$$

winds are assumed to carry specific angular momentum of the star

$$\longrightarrow \dot{J}_1 = h_1 \dot{M}_1$$

$$\dot{J}_{orb} = \dot{J}_1 + \dot{J}_2 = h_1 \dot{M}_1 + h_2 \cancelto{0}{\dot{M}_2}$$

$$\longrightarrow \left(\frac{\dot{J}}{J_{orb}}\right)_{wind} = \frac{h_1 \dot{M}_1}{J_{orb}} = \frac{J_{orb} M_2}{M_1 M_T} \frac{\dot{M}_1}{J_{orb}}$$

$$= \frac{M_2}{M_1} \frac{\dot{M}_1}{M_T} = \frac{1}{q_1} \frac{\dot{M}}{M_T}$$

$$\rightarrow \frac{\dot{a}}{a} = 2\left(\frac{\dot{J}}{J_{orb}}\right)_{wind} - \frac{2\dot{M}_1}{M_1} + \frac{\dot{M}_1}{M_T}$$

$$= \frac{2}{q_1}\frac{\dot{M}_1}{M_T} - \frac{2\dot{M}_1}{M_1} + \frac{\dot{M}_1}{M_T} = \frac{2M_2}{M_1}\frac{\dot{M}_1}{M_T} - \frac{2\dot{M}_1}{M_1}\frac{M_T}{M_T} + \frac{\dot{M}_1}{M_T}$$

$$\rightarrow \frac{\dot{a}}{a} = \frac{\dot{M}_1}{M_T}\left(\frac{2M_2 - 2M_T + M_1}{M_1}\right)$$

$$= \frac{\dot{M}_1}{M_T}\left(\frac{2M_2 - 2M_1 - 2M_2 + M_1}{M_1}\right)$$

$$= \frac{\dot{M}_1}{M_T}\left(\frac{-M_1}{M_1}\right) = -\frac{\dot{M}_1}{M_T}$$

$$\rightarrow \frac{\dot{a}}{a} = -\frac{\dot{M}_1}{M_T}$$

for a wind: $\dot{M}_1 < 0$

$\rightarrow \frac{\dot{a}}{a} > 0 \rightarrow$ orbit widens

Skip magnetic braking & GWS for now.

Roche lobe overflow:

Mass transfer can be driven by physics of the donor star or due to orbital response of the mass loss.

Donor star driving depends on how fast donor is evolving.

The mass loss is set by

$$\dot{M} = \frac{M_{donor}}{\tau} \rightarrow \tau: \text{dynamical, thermal, or nuclear}$$

$$\tau_{dyn} >> \tau_{therm} >> \tau_{nuc}$$

orbital response:

- mass loss or exchange
- tides
- magnetic braking
- GWs

Once the Roche Lobe is filled, the stability depends on how the radius of the star responds to losing mass:

$$\zeta \equiv \frac{d \log R}{d \log M} \quad \text{where } R(M) \sim M^{\zeta}$$

For MT to remain stable, the radius of the star can't significantly expand beyond the RL. Then

$$\zeta_* = \frac{d \log R_d}{d \log M_d} > \zeta_{RL} = \frac{d \log R_{RL}}{d \log M_d}$$

To determine when this is true, we can analytically calc ζ_{RL} and use stellar models to calc ζ_*

First: assume conservative mass transfer

Then:

$$\frac{\dot{a}}{a} = 2\frac{\dot{J}}{J} - \frac{2\dot{M}_d}{M_d} - 2\frac{\dot{M}_a}{M_a} + \frac{\dot{M}_T}{M_T}$$

for conservative mass transfer: $\dot{J} = 0 \equiv$ conserved

$\dot{M}_T = 0$

$\dot{M}_a = \dot{M}_d$

$$\Rightarrow \frac{\dot{a}}{a} = -2\frac{\dot{M}_d}{M_d} + 2\frac{\dot{M}_d}{M_a}$$

$$= -2\dot{M}_d\left(\frac{1}{M_d} - \frac{1}{M_a}\right) = -2\dot{M}_d\left(\frac{M_a - M_d}{M_a M_d}\right)$$

$$= 2\frac{\dot{M}_d}{M_d}(q-1) \quad q = M_d/M_a$$

* Can also write $\frac{\dot{M}_d}{M_d}$ in terms of q:

$$q = \frac{M_d}{M_a} \longrightarrow \dot{q} = \frac{\dot{M}_d}{M_a} - \frac{M_d}{M_a^2}\dot{M}_a$$

$$= \frac{\dot{M}_d}{M_a} + \frac{M_d}{M_a}\frac{\dot{M}_d}{M_a} = \frac{\dot{M}_d}{M_a}(1+q)\left(\frac{M_d}{M_d}\right)$$

$$= \frac{\dot{M}_d}{M_d}(1+q)\frac{M_d}{M_a} = \frac{\dot{M}_d}{M_d}q(1+q)$$

$$\rightarrow \frac{\dot{M}_d}{M_d} = \frac{\dot{q}}{q}\frac{1}{1+q} \text{ , then,}$$

$$\frac{\dot{a}}{a} = \frac{2\dot{M}_d}{M_d}(q-1) = \frac{2\dot{q}}{q}\frac{(q-1)}{(q+1)} = \frac{\dot{a}}{a}$$

* this is cool because we can look at what happens when $q < 1$ & $q > 1$

if $q < 1$: $\frac{M_d}{M_a} < 1 \rightarrow M_d < M_a$

if $q < 1$, $\dot{q} < 0$:

$$\frac{\dot{a}}{a} = (\text{negative})(\text{negative}) > 0$$

$\longrightarrow$ orbit widens!

if $q = 1$, $\dot{q} < 0$: $\frac{\dot{a}}{a} = 0 \longleftarrow$ unstable fixed point

if $q > 1$, $\dot{q} < 0$:

$$\frac{\dot{a}}{a} = (\text{negative})(\text{positive}) < 0$$

$\longrightarrow$ orbit shrinks!

We can determine stability by calculating ζ_{RL}

$$\zeta_{RL} = \frac{d \log R_{RL}}{d \log M_d}$$

$$R_{RL} = a\left[\frac{0.49 q^{2/3}}{0.6 q^{2/3} + \ln(1+q^{1/3})}\right]$$

$$\frac{\partial \ln R_{RL}}{\partial \ln M_d} = \frac{\partial \ln(a)}{\partial \ln(M_d)} + \frac{\partial \ln(R_{RL}/a)}{\partial \ln(q)} \frac{\partial \ln(q)}{\partial \ln M_d}$$

$$= \frac{2M_d^2 - 2M_a^2}{M_a(M_T)} + \left[\frac{2}{3} + \frac{q^{1/3}}{3} \frac{1.2 q^{1/3} + 1/(1+q^{1/3})}{0.6 q^{2/3} \quad \ln(1+q^{1/3})}\right] * [1+q]$$

→ if $\zeta_* > \zeta_{RL}$, then mass transfer proceeds stably. if not, you go into CE:

$$E_{bind,env} = \alpha_{CE} \Delta E_{orb} = \alpha_{CE}\left[\frac{G M_{1,f} M_2}{2a_f} - \frac{G M_{1,i} M_2}{2a_i}\right]$$

if $\alpha < 1$: energy is dissipated

if $\alpha > 1$: there is an energy source